Graham J. Buccheri

grahambuccheri@icloud.com +1 410-698-2761 in/gbuccheri grahambuccheri.github.io

EDUCATION

B.S. Computer Science

University of Maryland - College Park • College of Computer, Mathematical, and Natural Sciences | 2021 - 2025

CORE KNOWLEDGE AND SKILLS

Python, C++, C, C#, Ruby, Rust, Java, Ocaml, MIPS Assembly
Docker, Git, Gazebo/ROS, Unix, VirtualBox, Unity
Windows, Linux, Unix
XSS, CSRF, SQL injection, Cryptography Techniques, Secure Software Development, Software penetration testing, CTF Challenges
Machine Learning and Computer Vision: Computer Vision, Machine Learning, LLM, Roboflow, YOLOv5, OpenCV, NVIDIA CUDA
Software Techniques: Pathfinding Algorithms, SLAM (Simultaneous Localization and Mapping), geospatial data analysis, VR/Augmented Reality Development
Computer Networks, Network Security

EXPERIENCE

Software Developer

Trillium Engineering - Trillium Labs **February 2025 – Present**

- Ensured C++ and Python software complied with all flight safety and ITAR regulations
- Developed tools to dynamically manipulate video stream data for use in HIL simulations
- Improved Docker container usage to streamline builds and enhance deployment portability
- Accelerated Git-based Docker builds through optimized CI/CD pipeline integration

Computer Vision Software Developer

XFoundry@UMD **March 2024 – February 2025**

- Developed real-time wildfire detection algorithms for the XPRIZE Wildfire initiative, enhancing UAV efficiency for rapid response.
- Reduced fire detection latency by 1600% using CUDA-accelerated computer vision, leveraging GPU parallelism.
- Utilized deep learning models, such as YOLOv5, for real-time object detection, improving UAV wildfire surveillance.
- Spearheaded interdisciplinary team efforts to integrate computer vision and machine learning solutions, advancing UAV navigation, mapping, and situational awareness in complex environments.

Software Development/Aerospace Research

University of Maryland Department of Aerospace Engineering **March 2022 – August 2024**

- Applied Python and C++ to analyze and resolve critical flight system issues on unmanned aircraft, emphasizing advanced algorithm development and rigorous software testing to optimize performance.
- Engineered flight systems and image analysis software using OpenCV, integrating a Python-based obstacle avoidance system with Aruco tags to elevate obstacle detection and tracking capabilities.
- Utilized Python data grouping techniques to analyze shipping patterns from AIS data, integrating data structures, algorithms, and GeoJSON for STL anomaly detection, enhancing anomalous behavior identification.
- Performed flight tests and demonstrations to boost autopilot reliability and enhance autonomous flight safety.

Additive Manufacturing Lab Technical Supervisor

Terrapin Works **August 2023 – May 2024**

- Executed customer-specific manufacturing using additive manufacturing technologies (FDM, Resin, Nylon), ensuring precision and adherence to design specifications.
- Directed training programs for lab technicians, optimizing skills development and operational efficiency. Oversaw lab workflows, ensuring consistent quality of manufactured parts.
- Led development and production of aerodynamic components in collaboration with Formula SAE competition, enhancing consistency of component quality.
- Developed new processes to optimize GCODE files to improve part quality, reduce error rates, and minimize wasted materials.

Information Technology Internship

Baltimore County Public Schools **August 2019 – June 2021**

- Maintain network systems in order to streamline processes.
- Conducted comprehensive audits of security vulnerabilities within local networks.
- Collaborated with the technology team on various projects, demonstrating problem-solving skills and a passion for technology by learning new software and tackling challenging tech problems, thereby fostering a culture of continuous improvement and innovation.

PROJECTS

Game Developer

Independent • 2022 – 2024

- Contributed to the development of a submarine simulation in C#, implementing with Unity Game Engine, focusing on procedural terrain generation.
- Optimized terrain generation algorithms, enhancing game performance by improved frame rates on standard hardware, showcasing problem-solving and reliability in technology implementation.
- Engaged in continuous learning and application of new technologies, including data structures and software testing, to ensure the game's optimization and reliability.
- Collaborated with peers in algorithm development and testing phases, demonstrating teamwork and a passion for technology, leading to a more robust and engaging game experience.